

Firm Level Variation Analysis

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Libraries

```
library(here)
library(readr)
library(dplyr)
library(ggplot2)
library(lubridate)
library(tidyr)
```

Load data

```
# Load FDA approval data

fda_approvals <- read_csv(
  here(
    "processed_data/fda_approvals/fda_sdud_pdl_merged.csv"
  )
) |>
rename(
  approvals = n_approvals,
  firm = sponsor_name
) |>
mutate(post = approval_year >= 2014) |>
filter(approval_year >= 2010)

# Load patent data

patents <- read_csv(
  here(
    "processed_data/patents/patents_sdud_pdl_merged.csv"
  )
) |>
rename(
  atc_level1 = ATC1,
  patents = num_patents) |>
mutate(post = patent_year >= 2014) |>
filter(patent_year >= 2010)
```

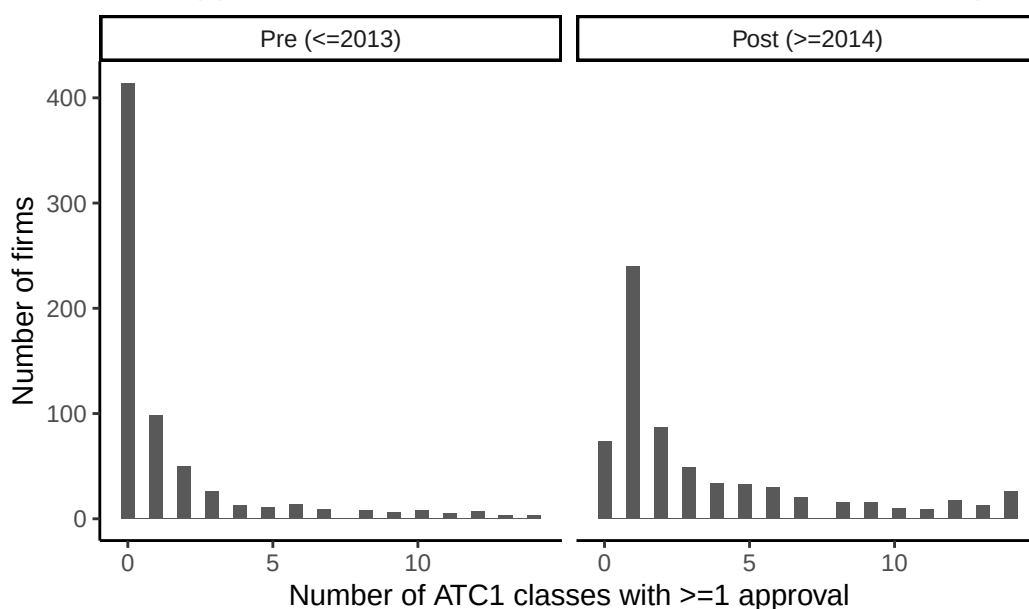
Between firm variation for ATCs

FDA

```
fda_firm_atc_prepost <- fda_approvals |>
  group_by(firm, atc_level1, post) |>
  summarise(any_innov = max(approvals > 0), .groups = "drop") |>
  group_by(firm, post) |>
  summarise(
    n_atc_innov = sum(any_innov),
    .groups = "drop"
  )

# Plot: FDA
ggplot(fda_firm_atc_prepost, aes(x = n_atc_innov)) +
  geom_histogram(bins = 30) +
  facet_wrap(~ post, labeller = as_labeller(c(`FALSE` = "Pre ( 2013)", `TRUE` = "Post ( 2014)"))
  labs(
    title = "FDA approvals: number of ATC1 classes with innovation (by firm)",
    x = "Number of ATC1 classes with 1 approval",
    y = "Number of firms"
  ) +
  theme_classic()
```

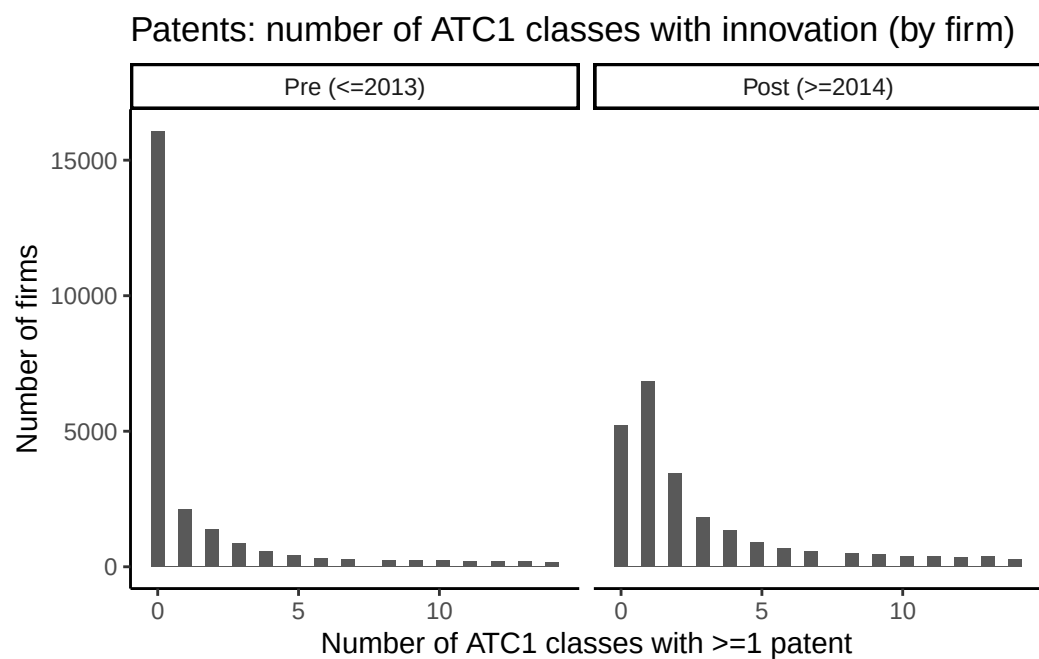
FDA approvals: number of ATC1 classes with innovation (by firm)



Patents

```
patent_firm_atc_prepost <- patents |>
  group_by(firm, atc_level1, post) |>
  summarise(any_innov = max(patents > 0), .groups = "drop") |>
  group_by(firm, post) |>
  summarise(
    n_atc_innov = sum(any_innov),
    .groups = "drop"
  )

ggplot(patent_firm_atc_prepost, aes(x = n_atc_innov)) +
  geom_histogram(bins = 30) +
  facet_wrap(~ post, labeller = as_labeller(c(`FALSE` = "Pre ( 2013)", `TRUE` = "Post ( 2014)"))
  labs(
    title = "Patents: number of ATC1 classes with innovation (by firm)",
    x = "Number of ATC1 classes with 1 patent",
    y = "Number of firms"
  ) +
  theme_classic()
```



Within firm variation

FDA

```
#| results: asis
#| fig-width: 10
#| fig-height: 6

fda_firm_atc_counts <- fda_approvals |>
  group_by(firm, atc_level1, post) |>
  summarise(approvals = sum(approvals, na.rm = TRUE), .groups = "drop") |>
  filter(approvals > 0)

fda_firm_index <- fda_firm_atc_counts |>
  group_by(firm) |>
  summarise(total_approvals = sum(approvals), .groups = "drop") |>
  arrange(desc(total_approvals)) |>
  mutate(firm_id = row_number())

fda_plot_data <- fda_firm_atc_counts |>
  left_join(fda_firm_index, by = "firm") |>
  pivot_wider(names_from = post, values_from = approvals, values_fill = 0) |>
  rename(pre = `FALSE`, post_val = `TRUE`)

atc_list_fda <- unique(fda_plot_data$atc_level1)

for (i in atc_list_fda) {

  # Filter for the ATC and apply the "Both > 5" rule
  single_atc_fda <- fda_plot_data |>
    filter(atc_level1 == i) |>
    filter(pre >= 1 & post_val >= 1) # The Volume Filter

  # Skip the plot if no firms meet this strict criteria
  if(nrow(single_atc_fda) == 0) {
    cat("\n\n### ATC Level", i, ": No firms meet the >5 approvals threshold in both periods.\n")
    next
  }

  p <- ggplot(single_atc_fda, aes(x = firm_id)) +
```

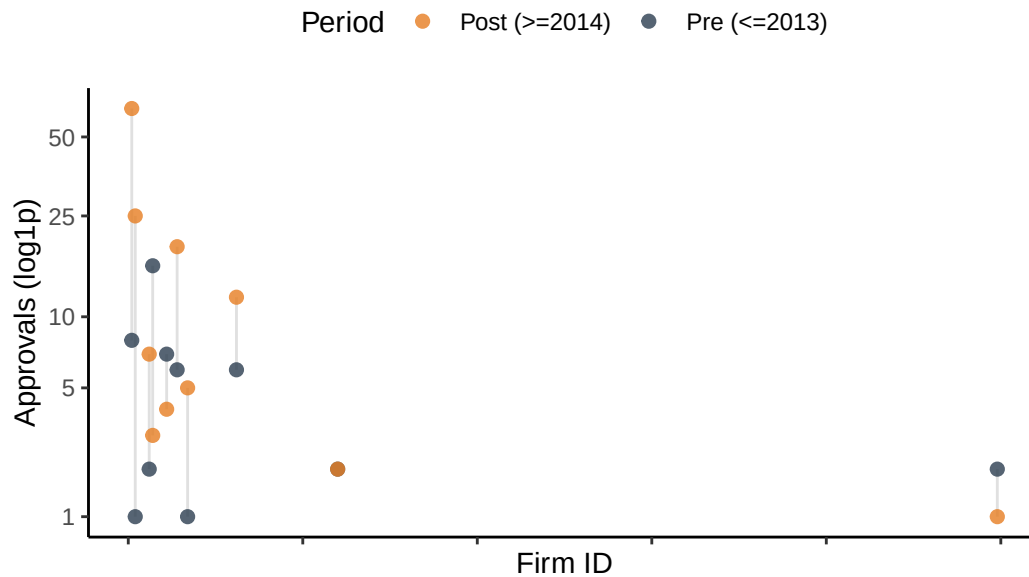
```

geom_segment(aes(xend = firm_id, y = pre, yend = post_val),
             color = "grey80", alpha = 0.6) +
geom_point(aes(y = pre, color = "Pre (<=2013)"), size = 2, alpha = 0.8) +
geom_point(aes(y = post_val, color = "Post (>=2014)"), size = 2, alpha = 0.8) +
scale_color_manual(values = c("Pre (<=2013)" = "#2c3e50", "Post (>=2014)" = "#e67e22")) +
scale_y_continuous(trans = "log1p", breaks = c(1, 5, 10, 25, 50, 100, 250, 500)) +
labs(title = paste("FDA High-Volume Firms (>1): ATC Level P", i),
     x = "Firm ID", y = "Approvals (log1p)", color = "Period") +
theme_classic() +
theme(axis.text.x = element_blank(), legend.position = "top")

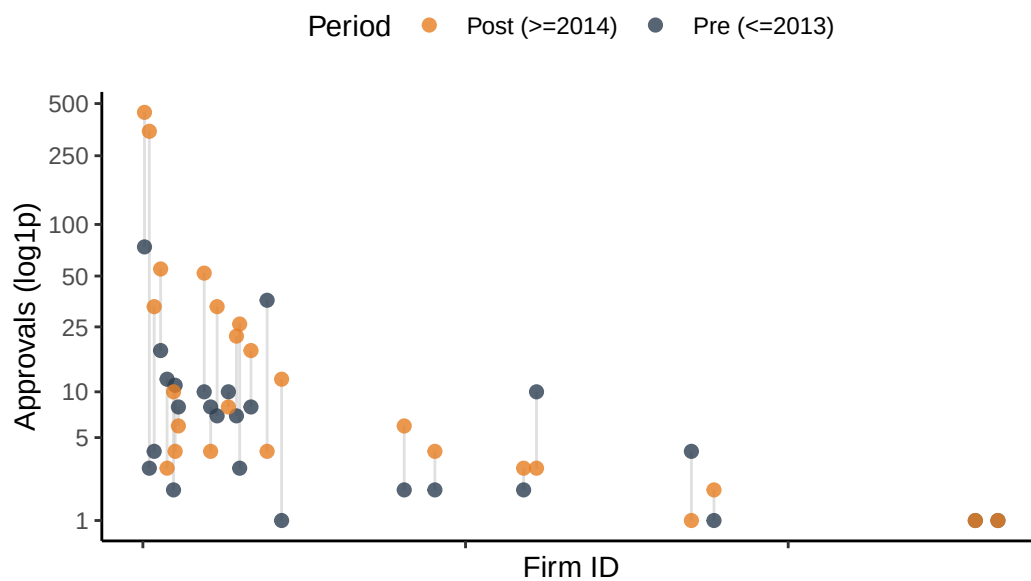
print(p)
}

```

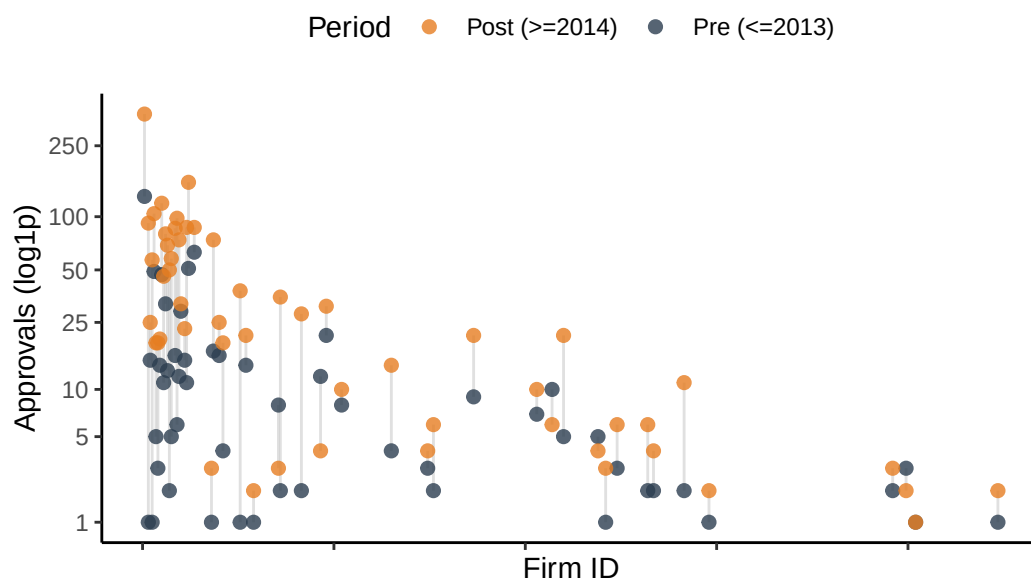
FDA High-Volume Firms (>1): ATC Level P



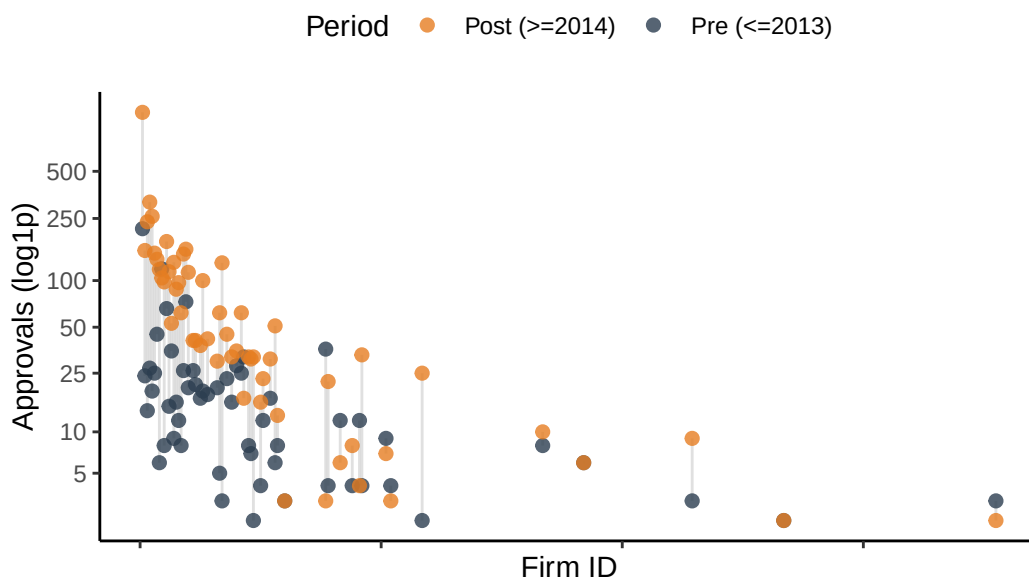
FDA High-Volume Firms (>1): ATC Level V



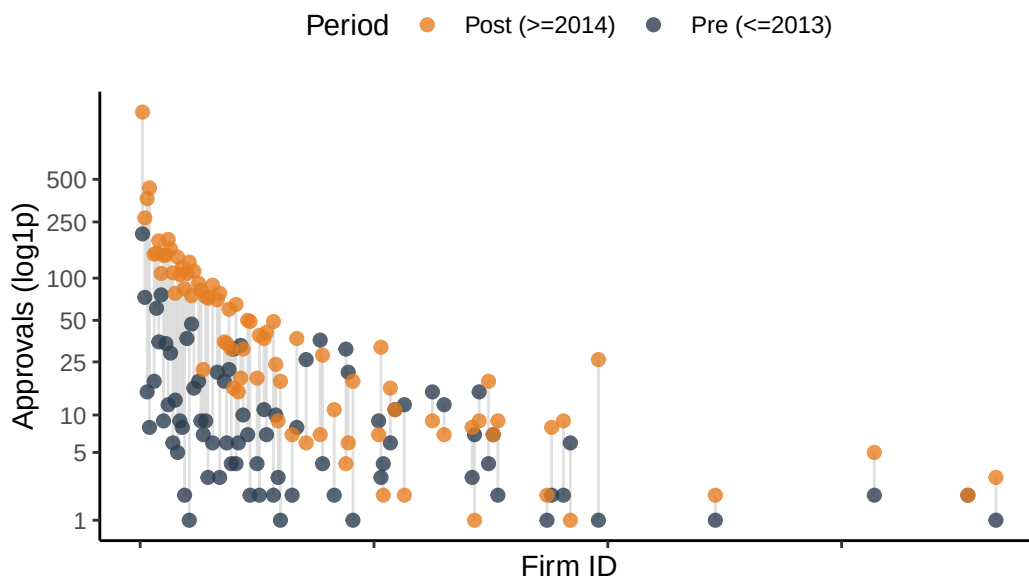
FDA High-Volume Firms (>1): ATC Level L



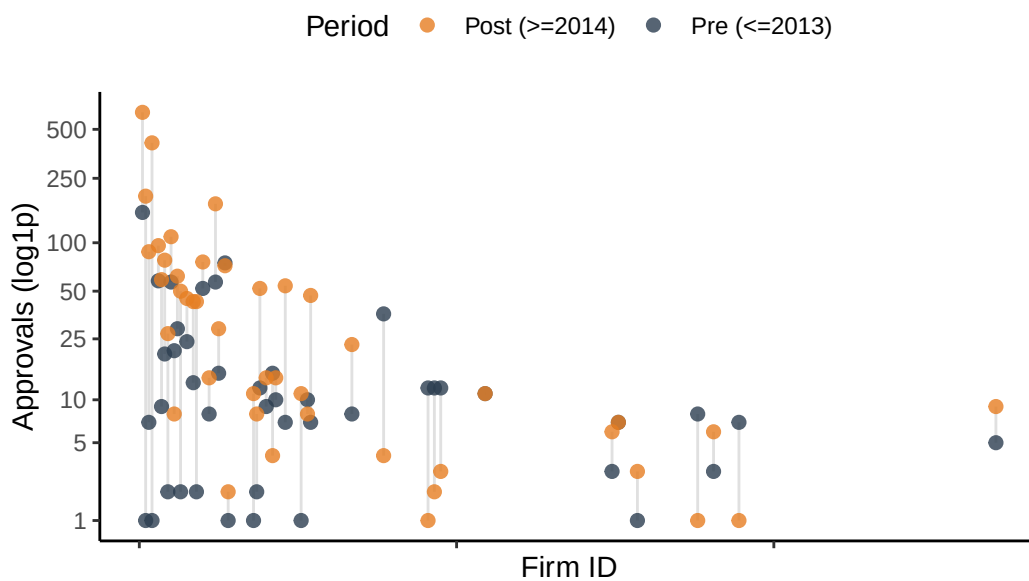
FDA High-Volume Firms (>1): ATC Level S



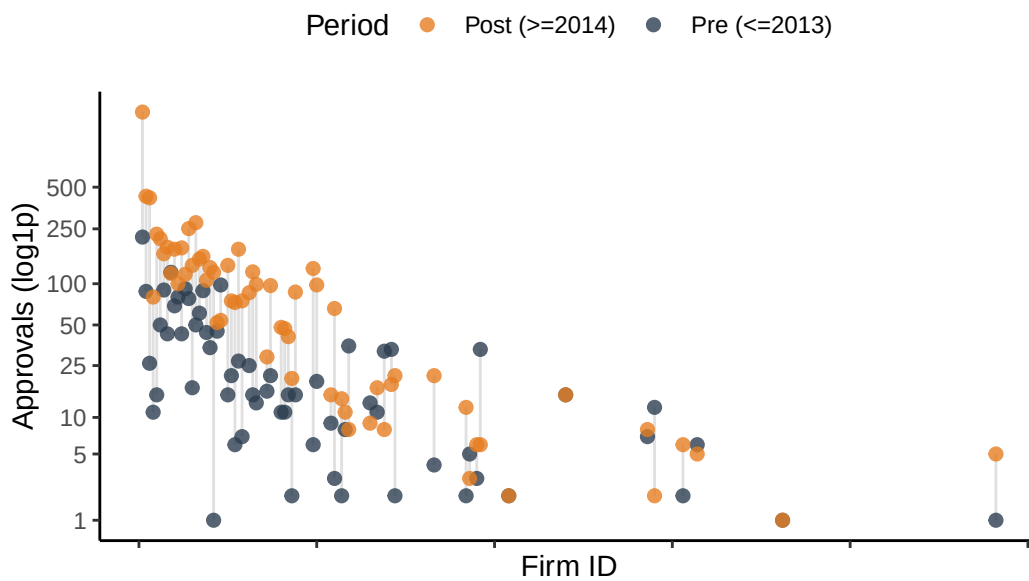
FDA High-Volume Firms (>1): ATC Level A



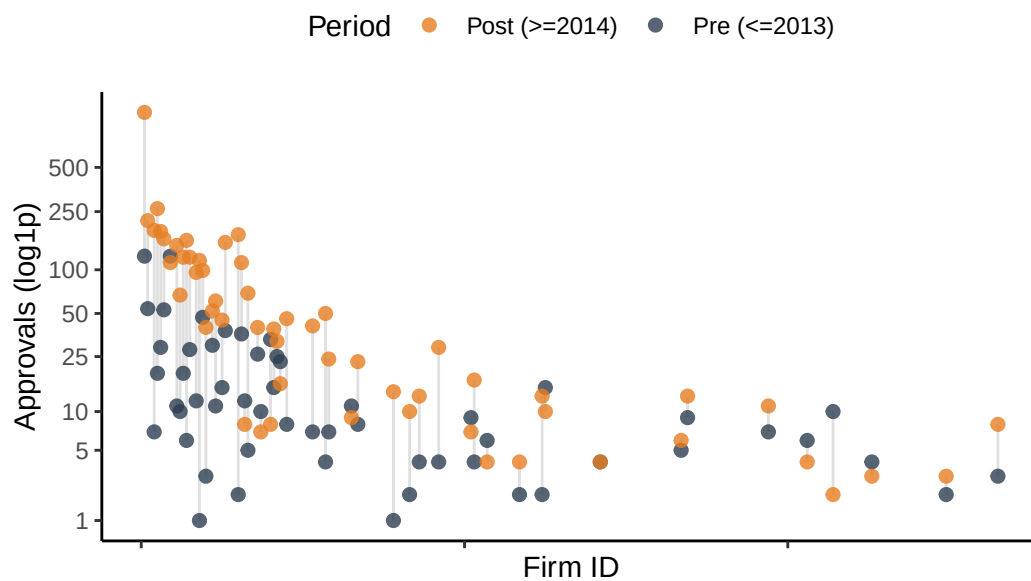
FDA High-Volume Firms (>1): ATC Level B



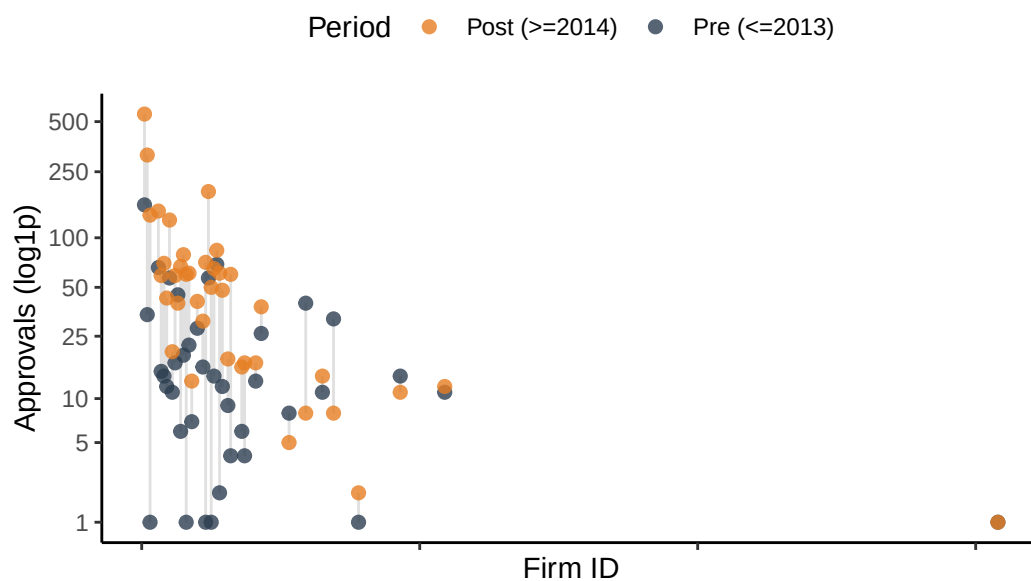
FDA High-Volume Firms (>1): ATC Level C



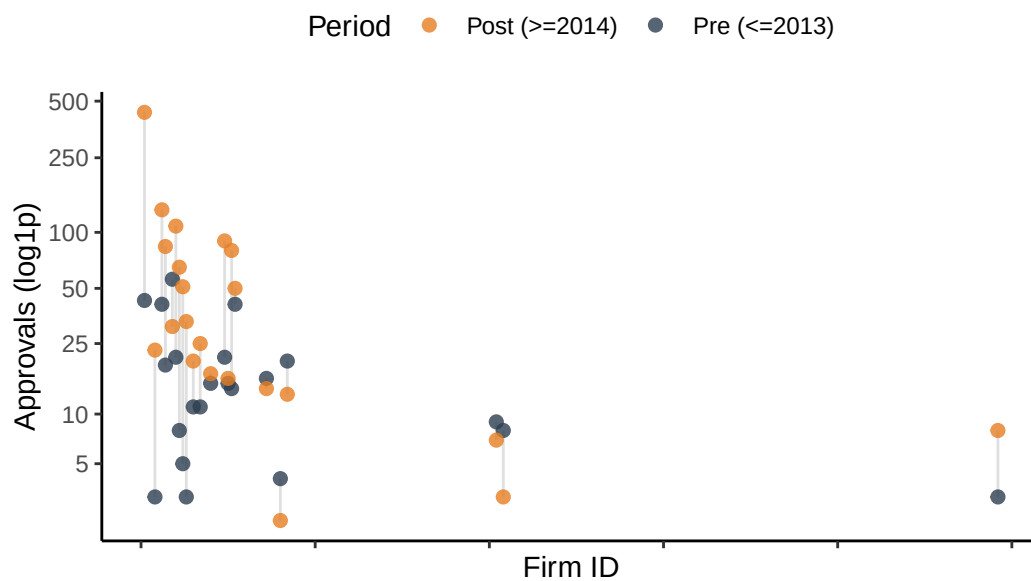
FDA High-Volume Firms (>1): ATC Level D



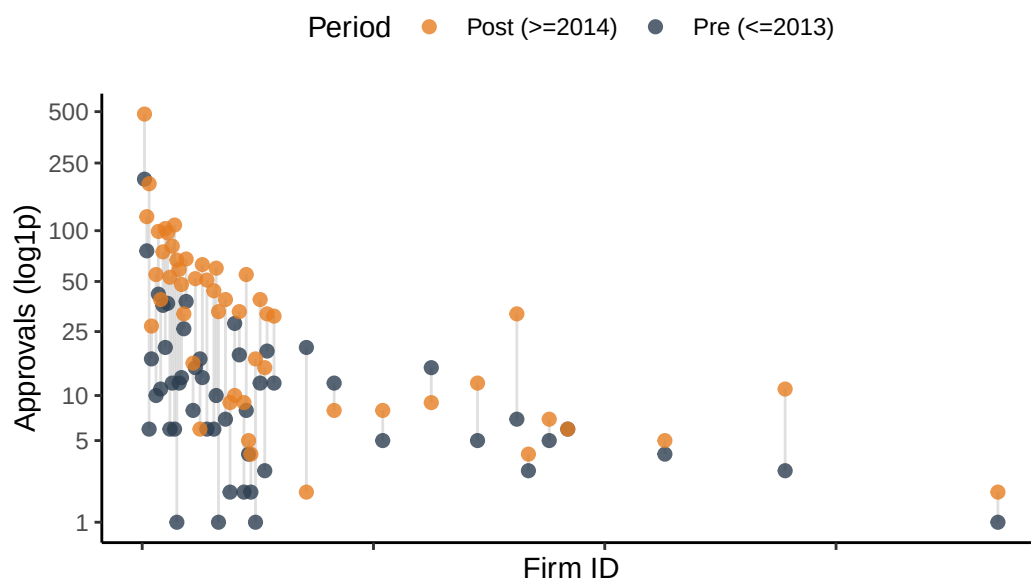
FDA High-Volume Firms (>1): ATC Level G



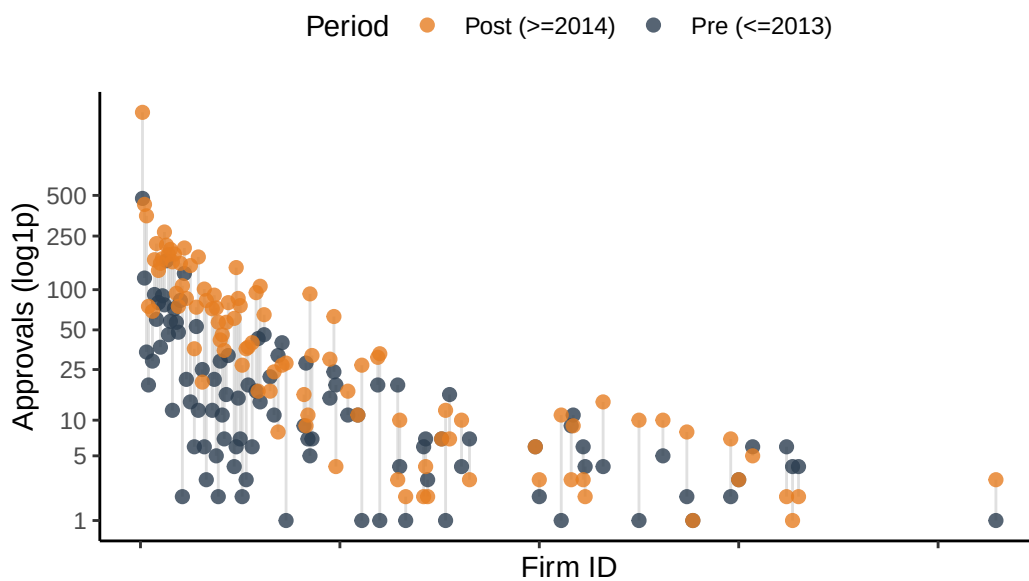
FDA High-Volume Firms (>1): ATC Level H



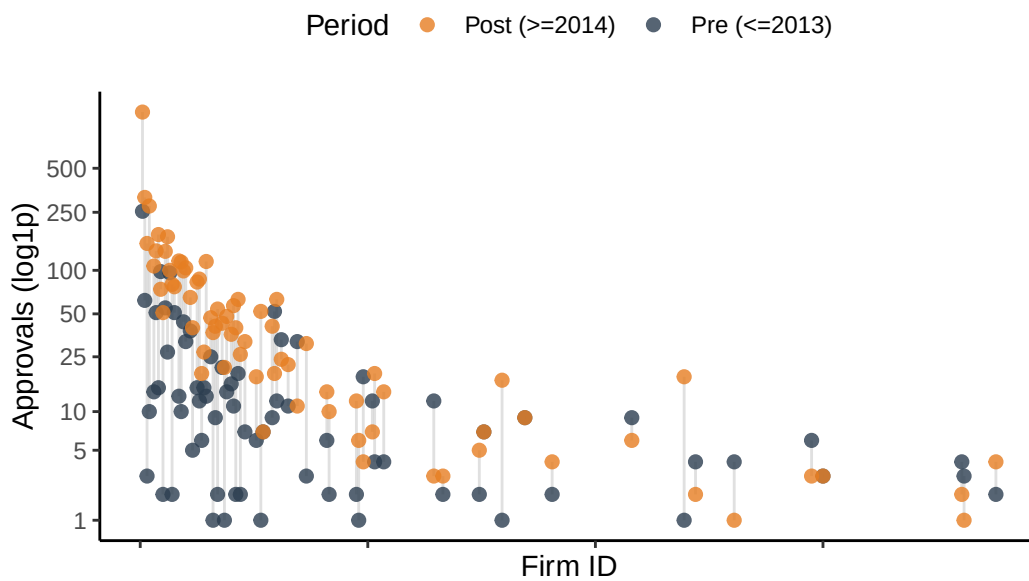
FDA High-Volume Firms (>1): ATC Level J



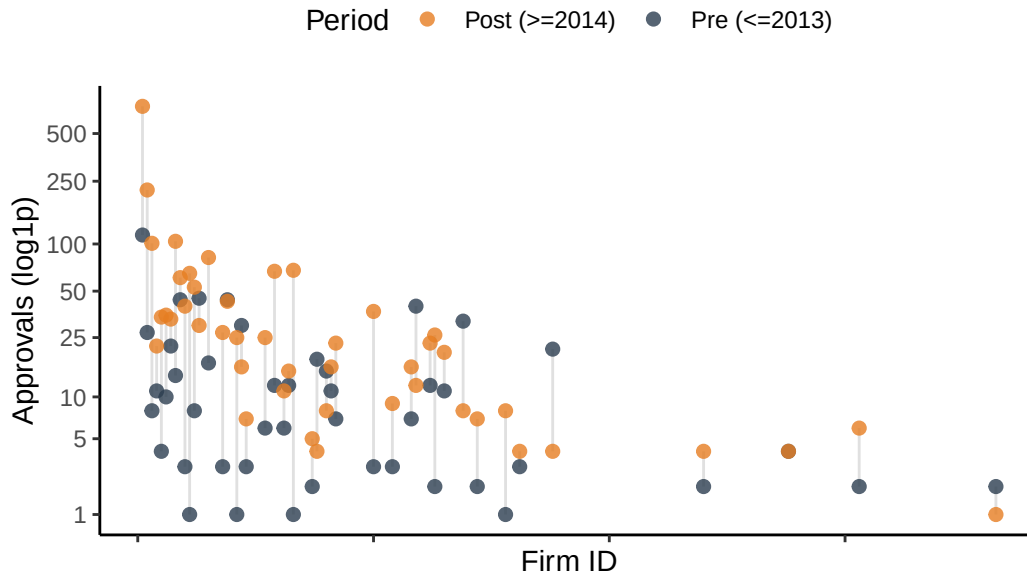
FDA High-Volume Firms (>1): ATC Level N



FDA High-Volume Firms (>1): ATC Level R



FDA High-Volume Firms (>1): ATC Level M



Patents

```
#| results: asis
#| fig-width: 10
#| fig-height: 6

patent_firm_atc_counts <- patents |>
  group_by(firm, atc_level1, post) |>
  summarise(patents = sum(patents, na.rm = TRUE), .groups = "drop") |>
  filter(patents > 0)

firm_index <- patent_firm_atc_counts |>
  group_by(firm) |>
  summarise(total_activity = sum(patents), .groups = "drop") |>
  arrange(desc(total_activity)) |>
  mutate(firm_id = row_number())

patent_plot_data <- patent_firm_atc_counts |>
  left_join(firm_index, by = "firm") |>
  pivot_wider(names_from = post, values_from = patents, values_fill = 0) |>
  rename(pre = `FALSE`, post_val = `TRUE`)
```

```

atc_list <- unique(patent_plot_data$atc_level1)

for (i in atc_list) {

  single_atc_data <- patent_plot_data |>
    filter(atc_level1 == i) |>
    filter(pre >= 1 & post_val >= 1) # The Volume Filter

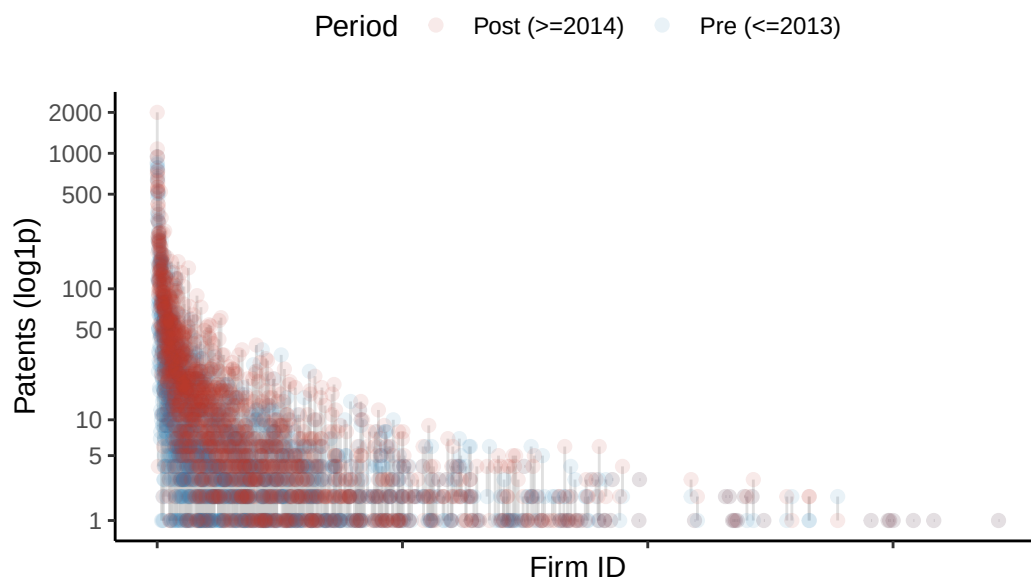
  if(nrow(single_atc_data) == 0) next

  p <- ggplot(single_atc_data, aes(x = firm_id)) +
    geom_segment(aes(xend = firm_id, y = pre, yend = post_val),
      color = "grey80", alpha = 0.6) +
    geom_point(aes(y = pre, color = "Pre (<=2013)"), size = 2, alpha = 0.1) +
    geom_point(aes(y = post_val, color = "Post (>=2014)"), size = 2, alpha = 0.1) +
    scale_color_manual(values = c("Pre (<=2013)" = "#2980b9", "Post (>=2014)" = "#c0392b")) +
    scale_y_continuous(trans = "log1p", breaks = c(1, 5, 10, 50, 100, 500, 1000, 2000)) +
    labs(title = paste("Patent High-Volume Firms (>1): ATC Level", i),
      x = "Firm ID", y = "Patents (log1p)", color = "Period") +
    theme_classic() +
    theme(axis.text.x = element_blank(), legend.position = "top")

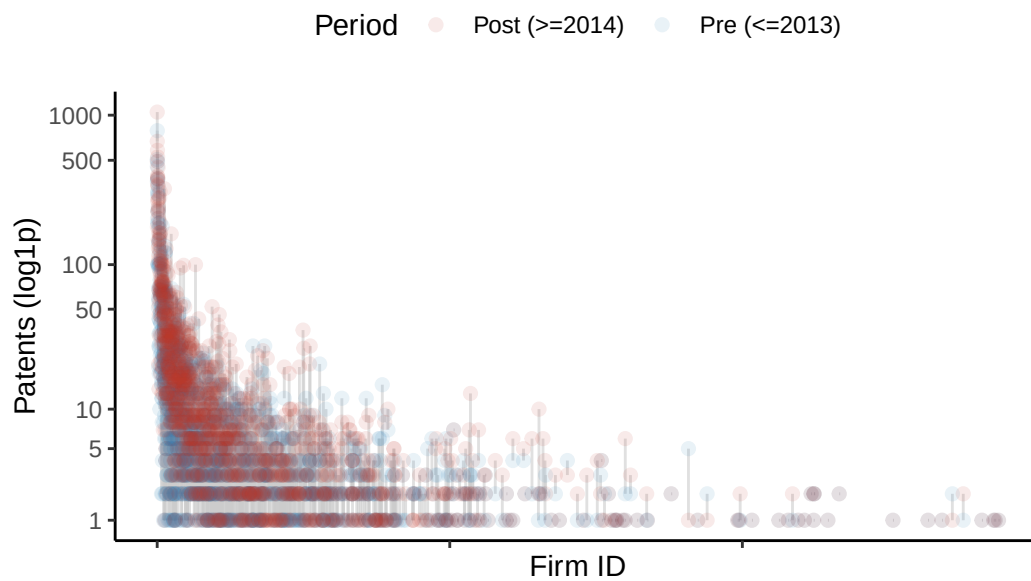
  print(p)
}

```

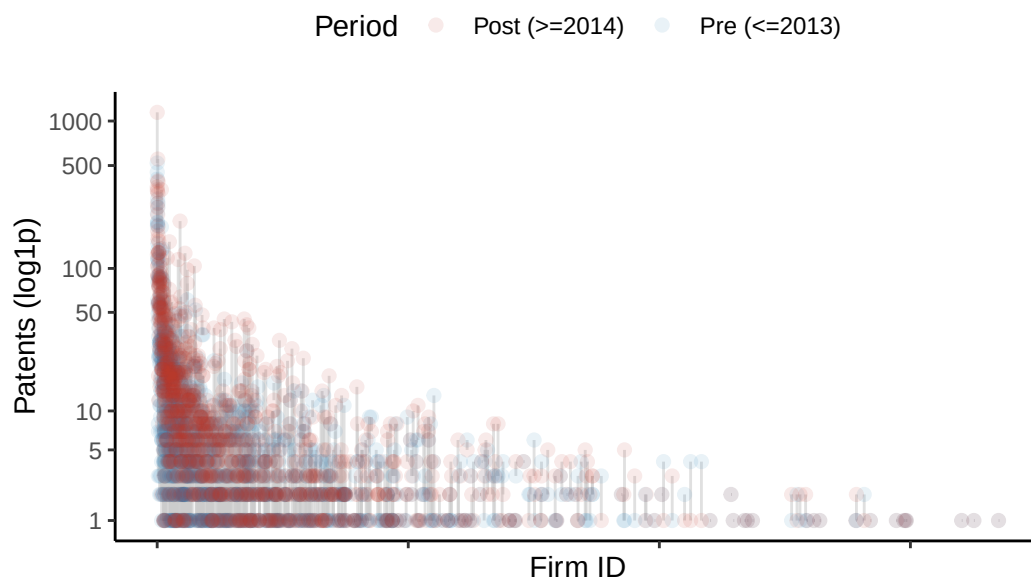
Patent High-Volume Firms (>1): ATC Level A



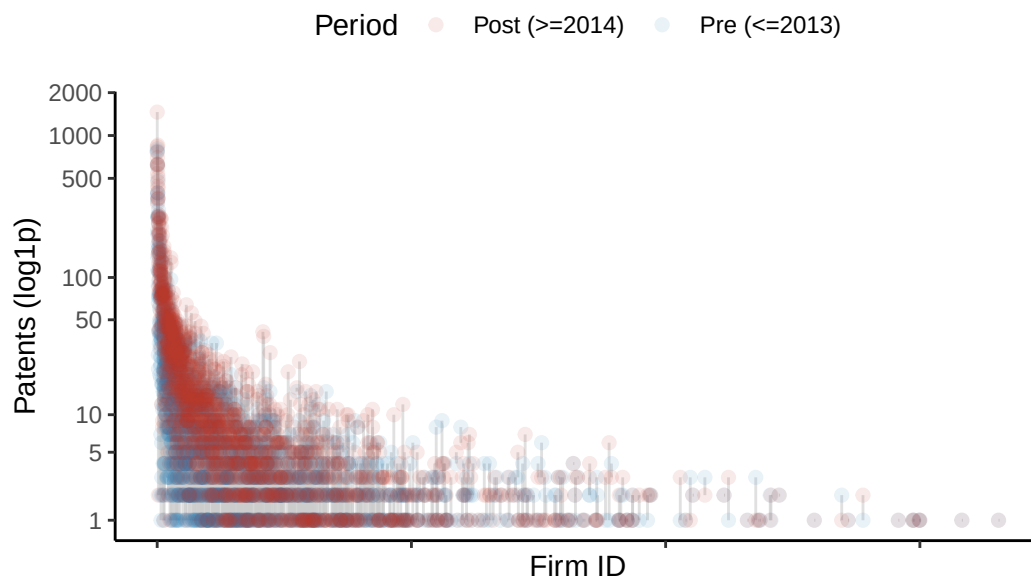
Patent High-Volume Firms (>1): ATC Level C



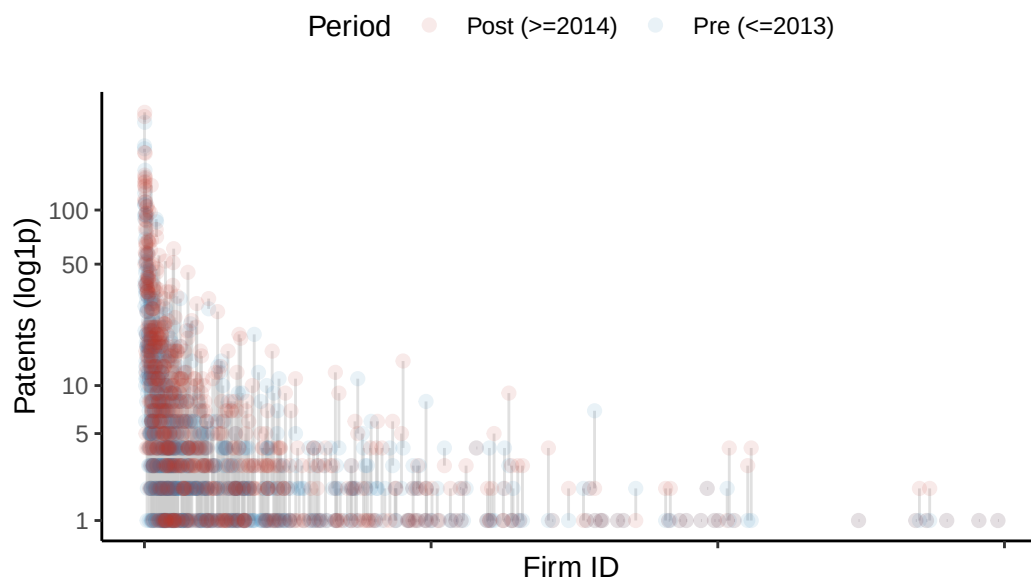
Patent High-Volume Firms (>1): ATC Level D



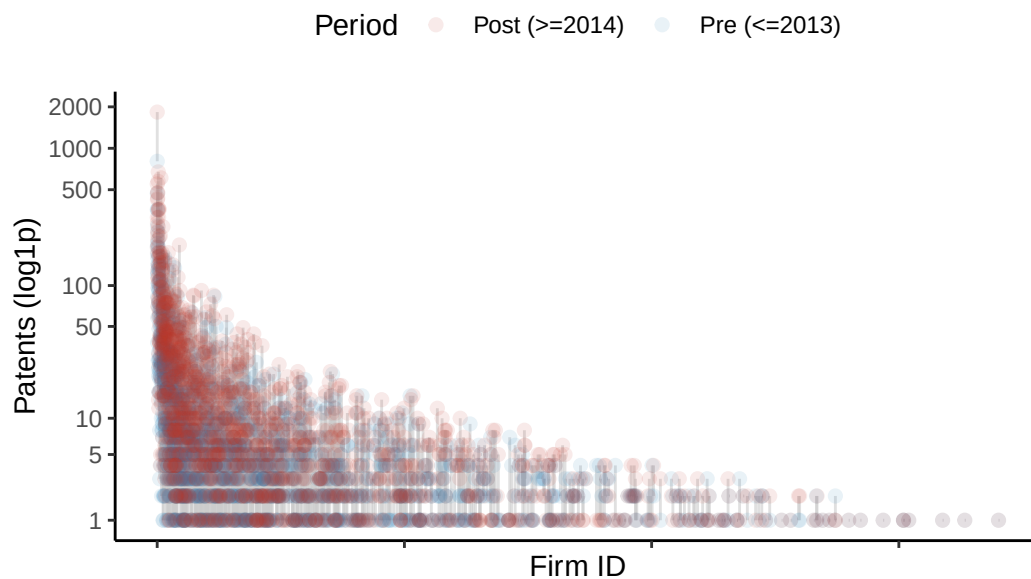
Patent High-Volume Firms (>1): ATC Level M



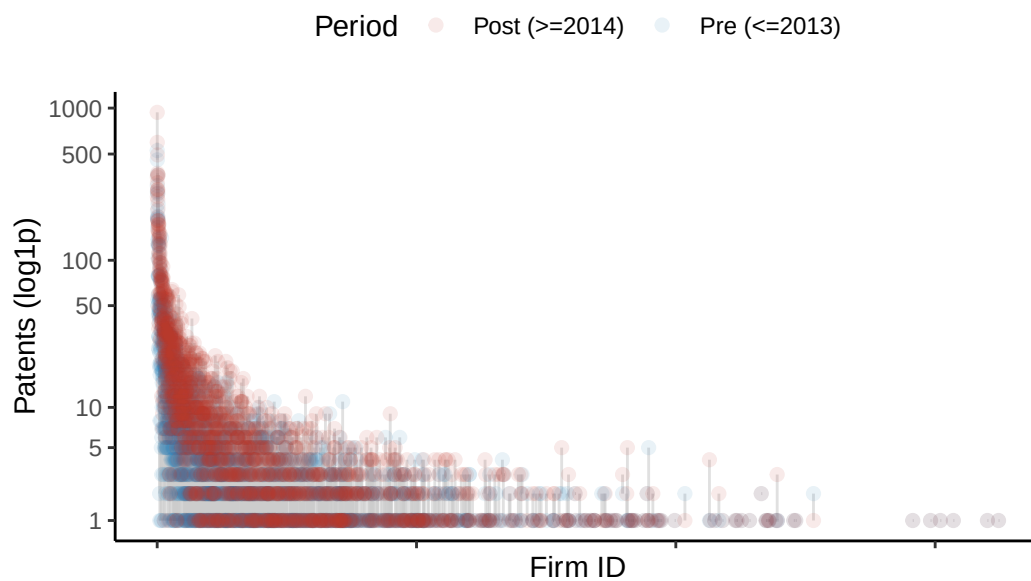
Patent High-Volume Firms (>1): ATC Level B



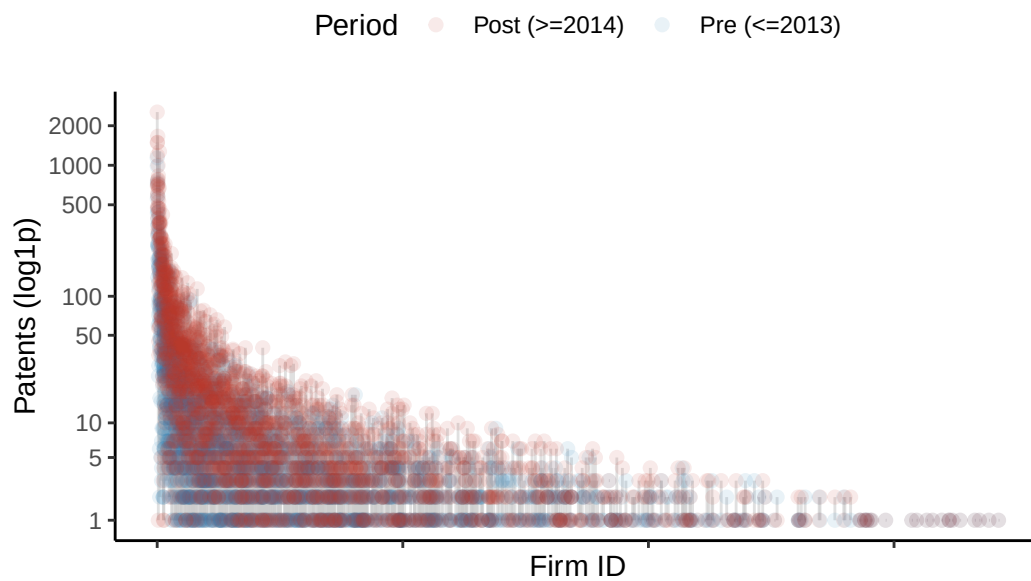
Patent High-Volume Firms (>1): ATC Level J



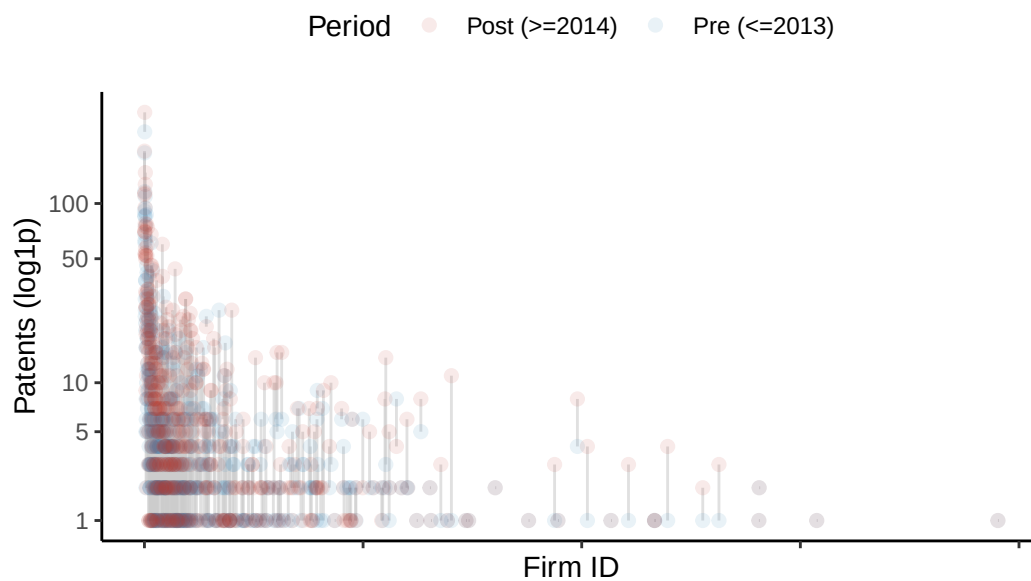
Patent High-Volume Firms (>1): ATC Level V



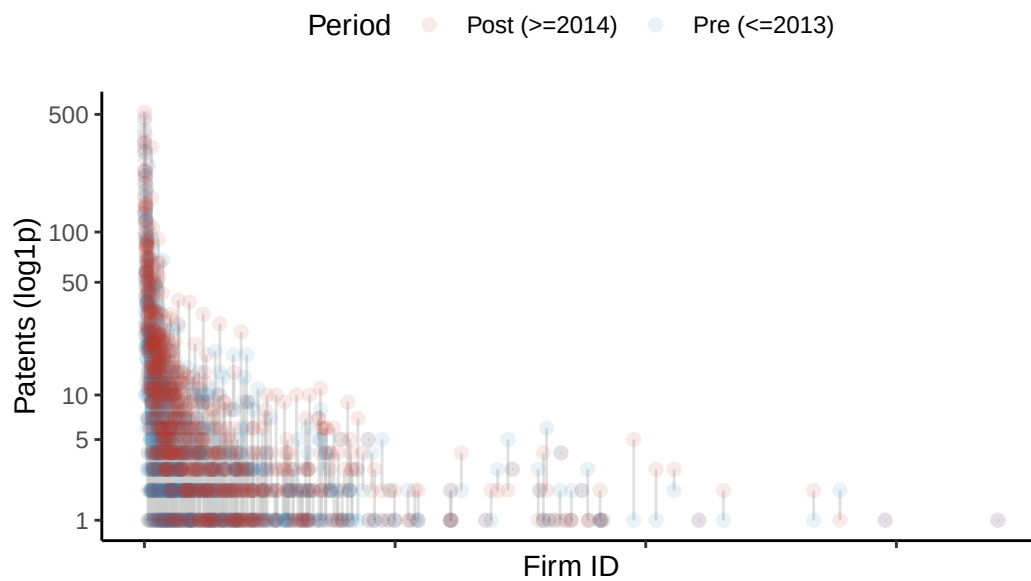
Patent High-Volume Firms (>1): ATC Level L



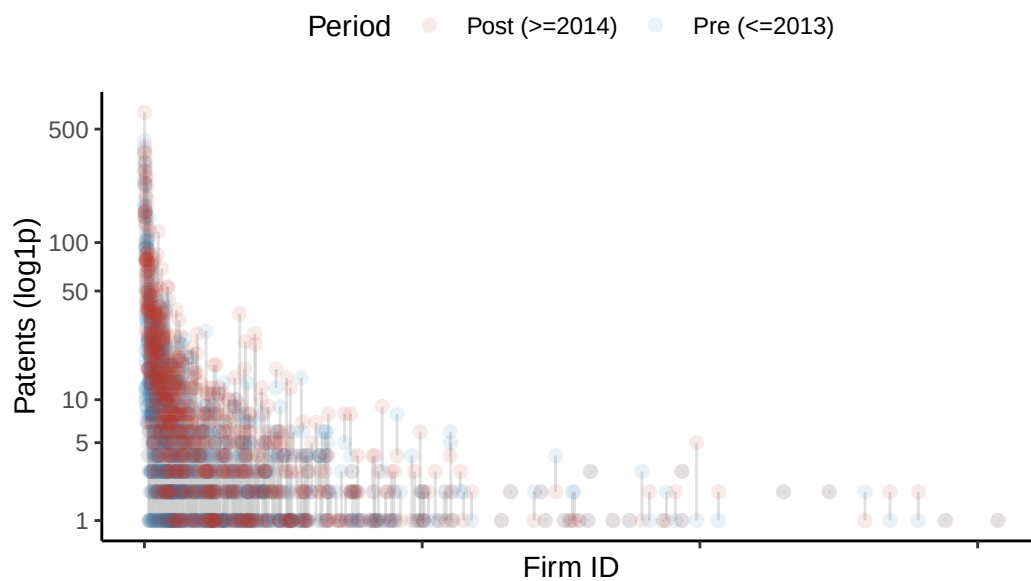
Patent High-Volume Firms (>1): ATC Level H



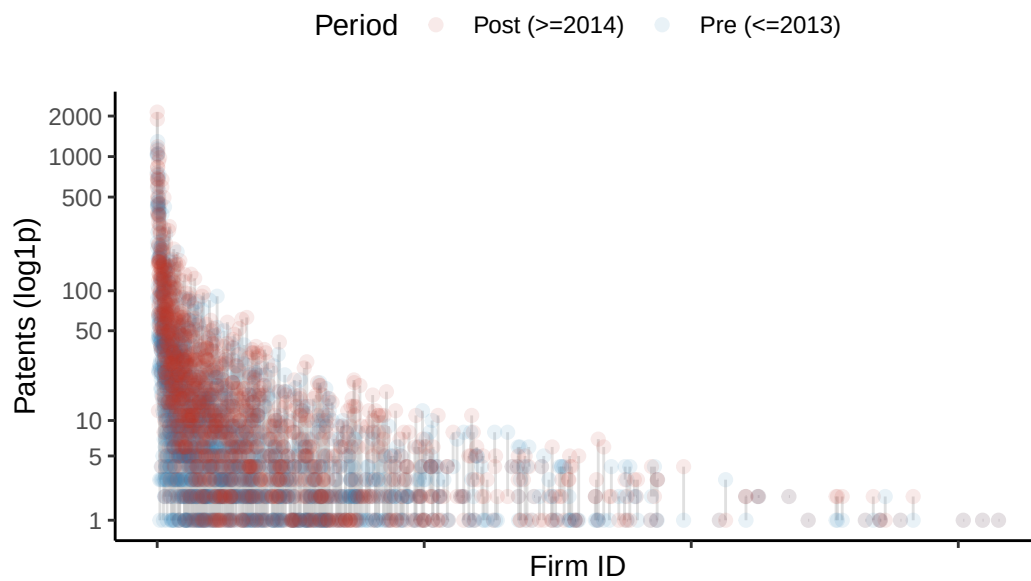
Patent High-Volume Firms (>1): ATC Level R



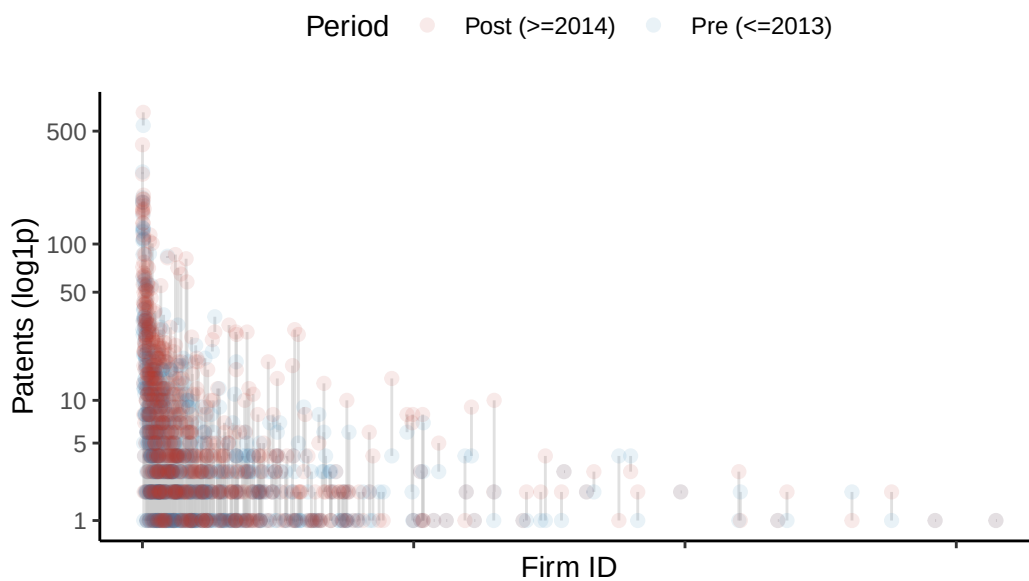
Patent High-Volume Firms (>1): ATC Level G



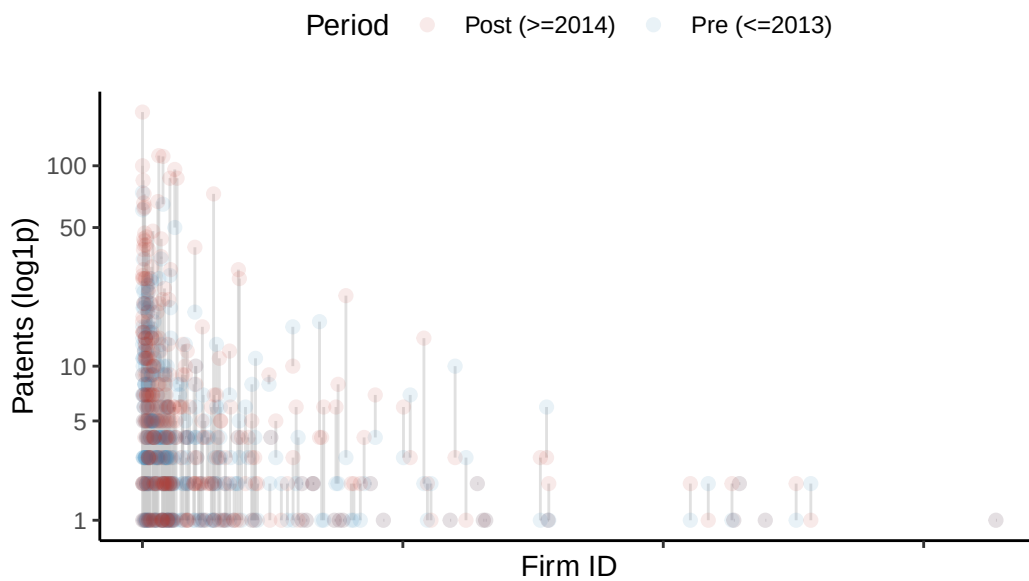
Patent High-Volume Firms (>1): ATC Level N



Patent High-Volume Firms (>1): ATC Level S



Patent High-Volume Firms (>1): ATC Level P



Variance analysis

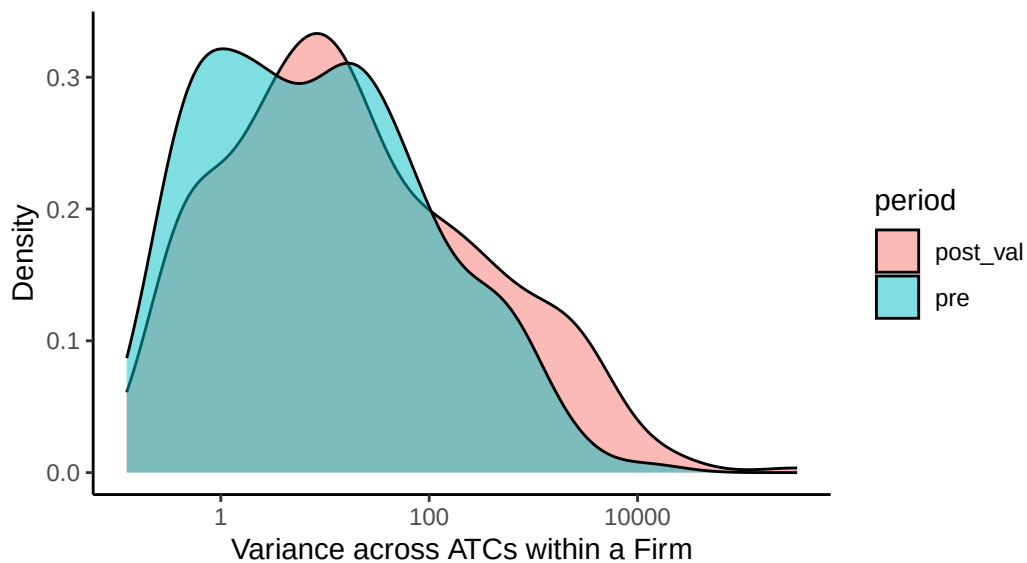
FDA

```
fda_firm_atc_variance <- fda_plot_data |>
  # Pivot to long to get one row per firm-atc-period
  pivot_longer(cols = c(pre, post_val), names_to = "period", values_to = "count") |>
  group_by(firm, period) |>
  # Only keep firms active in more than 1 ATC in that period
  filter(n_distinct(atc_level1) > 1) |>
  summarise(
    atc_variance = var(count),
    n_atcs = n_distinct(atc_level1),
    .groups = "drop"
  ) |>
  filter(atc_variance > 0)

# FDA Variance Distribution
ggplot(fda_firm_atc_variance, aes(x = atc_variance, fill = period)) +
  geom_density(alpha = 0.5) +
  scale_x_log10() +
  labs(
    title = "Portfolio Variation: FDA Approvals (Across ATCs)",
    subtitle = "Higher variance = more 'uneven' innovation across therapeutic classes",
    x = "Variance across ATCs within a Firm",
    y = "Density"
  ) +
  theme_classic()
```

Portfolio Variation: FDA Approvals (Across ATCs)

Higher variance = more 'uneven' innovation across therapeutic classes



Patents

```
patent_firm_atc_variance <- patent_plot_data |>
  pivot_longer(cols = c(pre, post_val), names_to = "period", values_to = "count") |>
  group_by(firm, period) |>
  filter(n_distinct(atc_level1) > 1) |>
  summarise(
    atc_variance = var(count),
    n_atcs = n_distinct(atc_level1),
    .groups = "drop"
  ) |>
  filter(atc_variance > 0)

# Patent Variance Distribution
ggplot(patent_firm_atc_variance, aes(x = atc_variance, fill = period)) +
  geom_density(alpha = 0.5) +
  scale_x_log10() +
  labs(
    title = "Portfolio Variation: Patents (Across ATCs)",
    x = "Variance across ATCs within a Firm",
    y = "Density"
```

```
) +  
theme_classic()
```

